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Multicategory Angle-Based Learning for Estimating Optimal Dynamic Treatment Regimes With Censored Data

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ABSTRACT

An optimal dynamic treatment regime (DTR) consists of a sequence of decision rules in maximizing long-term benefits, which is applicable for chronic diseases such as HIV infection or cancer. In this article, we develop a novel angle-based approach to search the optimal DTR under a multicategory treatment framework for survival data. The proposed method targets to maximize the conditional survival function of patients following a DTR. In contrast to most existing approaches which are designed to maximize the expected survival time under a binary treatment framework, the proposed method solves the multicategory treatment problem given multiple stages for censored data. Specifically, the proposed method obtains the optimal DTR via integrating estimations of decision rules at multiple stages into a single multicategory classification algorithm without imposing additional constraints, which is also more computationally efficient and robust. In theory, we establish Fisher consistency and provide the risk bound for the proposed estimator under regularity conditions. Our numerical studies show that the proposed method outperforms competing methods in terms of maximizing the conditional survival probability. We apply the proposed method to two real datasets: Framingham heart study data and acquired immunodeficiency syndrome clinical data. Supplementary materials for this article are available online.

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1. Introduction

Precision medicine tailored for individuals has become an important strategy in treating chronic diseases. Dynamic treatment regimes (DTRs) play a central role in precision medicine, such as recommendation of optimal treatments to individual patients according to patients' previous treatments and medical histories. In practice, an effective treatment strategy should not focus on short-term benefits, but aim for the most favorable long-term benefits. Consequently, the goal is to seek an optimal DTR, which is defined as a sequence of decision functions, to maximize the expected long-term benefits (Murphy 2003, 2005).

For chronic diseases such as cancer or HIV infection, survival data often arises, and developing an optimal DTR is critical under survival data framework. Recent developments of the optimal DTR estimation under survival data framework include Goldberg and Kosorok (2012), Zhao et al. (2014, 2018), and Hager, Tsiatis, and Davidian (2018), which maximize the expected survival time. However, in practice, maximizing the probability of surviving beyond a specific time could also be a primary concern, especially for patients with chronic diseases. For example, in cancer and HIV infection studies, a t -year survival probability is commonly practiced to evaluate survival risks or treatment effects (Lan, Laohasiriwong,

and Stewart 2013; Stefanie et al. 2015). To meet these clinical demands, we propose to maximize survival probability instead of the expected survival time in seeking an optimal DTR. Recent works focusing on maximization of survival probability include Bai, Tsiatis, and O'Brien (2013), Bai et al. (2017) and Jiang et al. (2017a, 2017b). Although these methods consider maximizing survival probability, they only studied one or two stages.

There are two main challenges in estimating the optimal DTR for censored data. First, treatment and covariate information from patients could be censoring in follow-up stages. Second, the true survival time might be unknown for patients who are still alive at the censoring time (Goldberg and Kosorok 2012). Methods proposed by Zhao et al. (2011), Jiang et al. (2017a), Bai et al. (2017), Huling and Yu (2018), Zhao et al. (2014), and Hager, Tsiatis, and Davidian (2018) only focus on one or two decision points (stages), while the censored Q-learning algorithm (Goldberg and Kosorok 2012) and the stabilized O-learning approach (Zhao et al. 2018) are able to deal with settings including more than two stages.

Nevertheless, none of the aforementioned methods can handle multicategory treatment scenarios when constructing a DTR. Two common approaches to solving multicategory classification problems are one-versus-one and one-versus-rest

approaches (Allwein, Schapire, and Singer 2000), which both apply sequential binary classifiers. However, the sequential binary classifier can yield a suboptimal result in some cases (Zhang and Liu 2013). Another approach to consider multiple treatment choices simultaneously is to estimate a classification function vector with a dimension determined by the number of categories (Vapnik and Vapnik 1998; Crammer and Singer 2001; Hastie et al. 2009; Liu and Yuan 2011). The corresponding decision rule assigns a subject to the category with the largest predicted element in the classification function vector for this subject. Typically, a sum-to-zero constraint, requiring the sum of elements from the classification function vector to be zero, is used to reduce the parameter space for desirable theoretical properties (Lee, Lin, and Wahba 2004; Liu and Yuan 2011). However, additional computational costs are incurred to solve the constrained optimization problem, which could be inefficient (Zhang and Liu 2014).

To overcome these drawbacks, Zhang and Liu (2014), Zhang et al. (2020), Fu et al. (2019), and Qi et al. (2020) proposed an angle-based large-margin classifier, which significantly reduces the computational cost without additional constraints, and outperforms other standard classification methods. Tao and Wang (2017) proposed an adaptive contrast weighted learning method to identify the optimal DTR. However, those approaches are neither applicable for multiple decision points nor can they handle censored outcomes.

In this article, we propose a new angle-based weighted approach for the optimal DTR estimation by maximizing the conditional survival probability under a multicategory treatment framework. Specifically, we propose a weighted Kaplan-Meier (KM) estimator for the survival function under a DTR through the inverse of the treatment probability at each stage. One key idea is to decompose the survival function at a given time-point to a product of survival probabilities at time-points before the given time. An advantage of this decomposition is that we only need to consider treatments received at stages before each of these previous time-points, since the survival probability at a certain time cannot be affected by treatments received later than that time. In this way, we can incorporate all available observations from patients into a weighted KM estimator, even though some patients may lack treatment and covariate information at certain stages due to censoring. In addition, we adopt an angle-based classifier to consider multiple potential treatment choices simultaneously and avoid the sum-to-zero constraint, which could be computationally restrictive in optimization.

Moreover, we propose to estimate decision rules for all stages simultaneously by maximizing the proposed objective function over a class of treatment regimes. In other words, our method integrates the estimations of decision rules at multiple stages as one joint weighted multicategory classification problem. In addition, the proposed method does not directly model the relationship between the survival outcomes and covariates or treatments, and thus mitigates the model-misspecification which may occur in Q-learning methods.

The main contributions of this article can be summarized as follows. First, to the best of our knowledge, this is the first work which directly estimates the optimal DTR under a

multicategory treatment framework for censored data. Second, the proposed method maximizes the survival probability in searching an optimal DTR with more than two decision stages. This has a profound impact in practice, since there is a high demand from long-term treatment management for making multistage decisions in treating chronic diseases. Third, the proposed method transforms DTR estimation problems at multiple stages into a joint multicategory classification problem and solves the optimization problem without imposing additional constraints, which improves computational efficiency and avoids potential overfitting problems. In theory, we establish Fisher consistency, which has not previously been established for the DTR estimation under the multicategory treatment framework.

The remainder of the article is organized as follows. Section 2 introduce the background and notation of the DTR and survival analysis. In Section 3, we propose a novel angle-based approach to estimate the optimal DTR under a multicategory treatment framework for survival data. Section 4 establishes the theoretical foundation of the proposed method. The computation algorithm is provided in Section 5. In Sections 6 and 7, we present performance of the proposed method through simulation studies and analyses of the Framingham Heart data and acquired immunodeficiency syndrome (AIDS) clinical data. Finally, we conclude with discussion in Section 8.

2. Background and Notations

In this section, we introduce background and notations for the standard DTR setting and survival analysis. We assume that the observed data are

$$\{(X_{i1}, A_{i1}, \dots, X_{im(Y_i)}, A_{im(Y_i)}, Y_i)\}_{i=1}^n,$$

containing n independent, identically distributed samples of $(X_1, A_1, \dots, X_{m(Y)}, A_{m(Y)}, Y)$, where $X_m \in \mathbb{R}^p$ denotes the covariates information collected between the $(m-1)$ th and m th stages, $A_m \in \mathcal{A}$ denotes the treatment assigned at the m th stage, $Y = T \wedge C$ denotes the observed survival time, T is a discrete survival time, and C is a censoring time. Without loss of generality, we assume that T takes values from $\{1, 2, \dots, G\}$. We also assume that $P(T = s)$ and $P(C = s)$ are both positive for each time-point $s = 1, \dots, G$ for the survival time T . Additionally, we use $m(Y)$ to represent the index of the stage where the time Y falls into, that is, either a subject is censored, or a failure event occurs at the stage $m(Y)$. There is a total of $m(G)$ stages. However, if $m(Y) < m(G)$, then the covariates and assigned treatments are not available for stages $m(Y) < m \leq m(G)$.

Note that each stage may contain multiple time-points, that is, $m(G)$ could be much smaller than G . We make a treatment decision at the beginning time-point of each stage. In particular, if there is only one time-point at each stage, then each s corresponds to a decision time-point for $1 \leq s \leq G$, which is equivalent to the interim decision analysis for discrete time-points (Armitage 1991). We let $\Delta = I(T \leq C)$ be an indicator for the occurrence of the failure event at or before the censoring time, and let $\tilde{A}_m = (A_1, \dots, A_m)^T$ and $\tilde{X}_m = (X_1^T, \dots, X_m^T)^T$ be longitudinal combinations of treatments and

covariates from the first stage to the m th stage, respectively. Let $\mathbf{H}_m = (\tilde{\mathbf{X}}_m^T, \tilde{\mathbf{A}}_{m-1}^T)^T$ denote accrued information up to the m th stage in which $\mathbf{H}_1 = \mathbf{X}_1$. Throughout this article, we make the following noninformative censoring assumption.

Assumption 1. The censoring time C is independent of survival time, covariates, and treatments.

For a given time-point g ($1 \leq g < G$), our goal is to find a sequence of decision rules $\tilde{\mathcal{D}}_{m_g}(\mathbf{H}_{m_g}) = \{\mathcal{D}_1(\mathbf{H}_1), \dots, \mathcal{D}_{m_g}(\mathbf{H}_{m_g})\}$ under which the survival function at g

$$\begin{aligned} S^{\tilde{\mathcal{D}}_{m_g}}(g) &= E_1 \left[E_2^{\tilde{\mathcal{D}}_1} \left[\dots E_{m_g}^{\tilde{\mathcal{D}}_{m_g-1}} \left[P\{T > g | \tilde{\mathbf{X}}_{m_g}, \tilde{\mathbf{A}}_{m_g} \right. \right. \right. \\ &\quad \left. \left. \left. = \tilde{\mathcal{D}}_{m_g}(\mathbf{H}_{m_g}) \right] \dots \right] \right] \\ &= E_1 \left[E_2^{\tilde{\mathcal{D}}_1} \left[\dots E_{m_g}^{\tilde{\mathcal{D}}_{m_g-1}} \left[E\{I(T > g) | \tilde{\mathbf{X}}_{m_g}, \tilde{\mathbf{A}}_{m_g} \right. \right. \right. \\ &\quad \left. \left. \left. = \tilde{\mathcal{D}}_{m_g}(\mathbf{H}_{m_g}) \right] \dots \right] \right], \end{aligned} \quad (1)$$

is maximized, E_1 and $E_m^{\tilde{\mathcal{D}}_{m-1}}$ represent expectations with respect to $\mathbf{X}_1$ and $\{\mathbf{X}_m | \tilde{\mathbf{X}}_{m-1}, \tilde{\mathbf{A}}_{m-1} = \tilde{\mathcal{D}}_{m-1}(\mathbf{H}_{m-1})\}$, respectively, for $2 \leq m \leq m_g$. Here, $m_g = m(g-1)$ since the survival probability at time g is assumed to be affected only by treatments at or before time $g-1$.

3. Angle-Based Weighted Kaplan–Meier Method

3.1. Survival Function Under a Sequence of Decision Rules

To estimate an optimal DTR which maximizes the $S^{\tilde{\mathcal{D}}_{m_g}}(g)$ in (1), we derive a weighted KM estimator for the survival function $S^{\tilde{\mathcal{D}}_{m_g}}(g)$ as follows. Let $E^{\tilde{\mathcal{D}}_{m_g}}$ denote the expectation with respect to distribution $\mathcal{P}^{\tilde{\mathcal{D}}_{m_g}}$, which is the conditional distribution of $(T, \mathbf{H}_{m_g}, \mathbf{A}_{m_g})$ under $\tilde{\mathbf{A}}_{m_g} = \tilde{\mathcal{D}}_{m_g}(\mathbf{H}_{m_g})$ with probability density function:

$$\begin{aligned} P(T = k | \mathbf{H}_{m_g} = \mathbf{h}_{m_g}, \mathbf{A}_{m_g} = \mathbf{a}_{m_g}) &= I\{\tilde{\mathbf{a}}_{m_g} = \tilde{\mathcal{D}}_{m_g}(\mathbf{h}_{m_g})\} \\ &\prod_{j=1}^{m_g} \psi_j(\mathbf{x}_j | \mathbf{h}_{j-1}, \mathbf{a}_{j-1}), \end{aligned} \quad (2)$$

where $1 \leq k \leq G$, $\psi_j(\mathbf{x}_j | \mathbf{h}_{j-1}, \mathbf{a}_{j-1})$ is the conditional density function of $\mathbf{X}_j$ given $\mathbf{H}_{j-1}$ and $\mathbf{A}_{j-1}$, and $\psi_1(\mathbf{x}_1 | \mathbf{h}_0, \mathbf{a}_0) = \psi_1(\mathbf{x}_1)$ is the unconditional density function of $\mathbf{X}_1$.

Since the censoring is random and able to occur at any time-point before g , we adopt the KM estimator to decompose the survival probability $S^{\tilde{\mathcal{D}}_{m_g}}(g)$ in terms of survival probabilities at time-points before g , and estimate $S^{\tilde{\mathcal{D}}_{m_g}}(g)$ as

$$S^{\tilde{\mathcal{D}}_{m_g}}(g) = E^{\tilde{\mathcal{D}}_{m_g}}[I(T > g)] = \prod_{s=1}^g q^{\tilde{\mathcal{D}}_{m_g}}(s), \quad (3)$$

where

$$q^{\tilde{\mathcal{D}}_{m_g}}(s) = \frac{E^{\tilde{\mathcal{D}}_{m_g}}[I(T \geq s+1)]}{E^{\tilde{\mathcal{D}}_{m_g}}[I(T \geq s)]}. \quad (4)$$

Under the multicategory framework, there are many potential combinations of assigned treatments $\tilde{\mathbf{A}}_{m_g}$, resulting only a few subjects satisfying $I\{\tilde{\mathbf{A}}_{m_g} = \tilde{\mathcal{D}}_{m_g}(\mathbf{H}_{m_g})\} = 1$ for the estimation of $q^{\tilde{\mathcal{D}}_{m_g}}(s)$. To use more subjects in estimation of $q^{\tilde{\mathcal{D}}_{m_g}}(s)$, we impose the following assumptions.

Assumption 2. For any $1 \leq s \leq g$, we assume that $I(T = s)$ and $I(T > s)$ are independent of $\mathbf{A}_{m(s)+1}$ given $\mathbf{H}_{m(s)+1}$.

Assumption 3. For each $1 \leq s \leq g$, we assume that

$$I(T = s), I(T > s) \perp\!\!\!\perp Z | \{\tilde{\mathbf{X}}_{m(s)+1}, \tilde{\mathbf{A}}_{m(s)} = \tilde{\mathcal{D}}_{m(s)}(\mathbf{H}_{m(s)})\}, \quad (5)$$

where Z represents $\mathbf{X}_m$ or $\mathbf{A}_m = \mathcal{D}_{m'}(\mathbf{H}_m)$ for $m(s)+1 < m \leq m_g$ and $m(s) < m' \leq m_g$.

Assumption 2 requires that all confounders of $I(T = s)$ (or $I(T > s)$) and $\mathbf{A}_{m(s)+1}$ are observed, and this assumption is similar to the unconfoundedness assumption (de Luna and Johansson 2014) which is standard in treatment effect literature. **Assumption 3** is based on the fact that the occurrence of a failure event at or after a specific time s can be affected only by the treatments and covariates before s . For $m > m(s)+1$ and $m' > m(s)$, $\mathbf{X}_m$ and $\mathbf{A}_{m'} = \mathcal{D}_{m'}(\mathbf{H}_{m'})$ may not be affected by $I(T = s)$ and $I(T > s)$ given $\tilde{\mathbf{X}}_{m(s)+1}$ and $\tilde{\mathbf{A}}_{m(s)} = \tilde{\mathcal{D}}_{m(s)}(\mathbf{H}_{m(s)})$. This is due to **Assumption 2** at Stage $m(s)+1$ and the fact that covariates at Stage $m(s)+1$ can capture information of $I(T = s)$ and $I(T > s)$ at the $m(s)$ th stage when a sufficient number of covariates are available.

Under **Assumption 3**, for each $1 \leq s \leq G$ such that $m(s) < m_g$, we can ignore the influence of the treatments after the $m(s)$ th stage, and the corresponding modified $q^{\tilde{\mathcal{D}}_{m_g}}(s)$ is

$$\begin{aligned} q^{\tilde{\mathcal{D}}_{m_g}}(s) &= 1 - \frac{E^{\tilde{\mathcal{D}}_{m_g}}[I(T = s)]}{E^{\tilde{\mathcal{D}}_{m_g}}[I(T \geq s)]} \\ &= 1 - \frac{E^{\tilde{\mathcal{D}}_{m(s)}}[I(T = s)]}{E^{\tilde{\mathcal{D}}_{m(s)}}[I(T \geq s)]}. \end{aligned} \quad (6)$$

The first equality in (6) is due to that T is discrete and only falls into $\{1, \dots, s, s+1, \dots, G\}$. Equation (6) for $q^{\tilde{\mathcal{D}}_{m_g}}(s)$ implies that decision rules after Stage $m(s)$ are irrelevant to the optimal decision rules which maximize survival probability at time-points in Stage $m(s)$. This distinguishes the proposed method from existing methods which consider the expected survival time as a target goal. For the estimation of the optimal decision rule at Stage $m(s)$, maximizing the expected survival time usually requires a counterfactual dataset where some subjects take the optimal treatments at stages after Stage $m(s)$. However, this requirement is not necessary under our framework. On the contrary, the proposed method is more robust, since the estimation of the optimal decision rule at any stage after Stage $m(s)$ does not affect the estimation at Stage $m(s)$.

However, samples from the distribution $\mathcal{P}^{\tilde{\mathcal{D}}_{m_g}}$ with the density in (2) are generally not observable, which implies that $E^{\tilde{\mathcal{D}}_{m(s)}}[I(T \geq s)]$ cannot be calculated based on the observed

data directly. To estimate $q^{\tilde{\mathcal{D}}_{m_g}}(s)$, we convert the expectation $E^{\tilde{\mathcal{D}}_{m(s)}}[I(T \geq s)]$ to an unconditional expectation using the Radon–Nikodym theorem. That is,

$$\begin{aligned} E^{\tilde{\mathcal{D}}_{m(s)}}[I(T \geq s)] &= \int I(T \geq s) d\mathcal{P}^{\tilde{\mathcal{D}}_{m(s)}} \\ &= \int I(T \geq s) \frac{d\mathcal{P}^{\tilde{\mathcal{D}}_{m(s)}}}{d\mathcal{P}_{m(s)}} d\mathcal{P}_{m(s)}, \end{aligned} \quad (7)$$

where $\mathcal{P}_{m(s)}$ denotes the unconditional distribution of $(T, \mathbf{H}_{m(s)}, A_{m(s)})$ with the following probability density function:

$$P(T = k | \mathbf{H}_{m(s)} = \mathbf{h}_{m(s)}, A_{m(s)} = a_{m(s)}) \prod_{j=1}^{m(s)} \pi_j(a_j | \mathbf{h}_j) \psi_j(\mathbf{x}_j | \mathbf{h}_{j-1}, a_{j-1}),$$

where $1 \leq k \leq G$ and $\pi_j(a_j | \mathbf{h}_j) = P(A_j = a_j | \mathbf{H}_j = \mathbf{h}_j)$. Since

$$\frac{d\mathcal{P}^{\tilde{\mathcal{D}}_{m(s)}}}{d\mathcal{P}_{m(s)}} = \frac{I[\tilde{\mathbf{A}}_{m(s)} = \tilde{\mathcal{D}}_{m(s)}(\mathbf{H}_{m(s)})]}{\prod_{j=1}^{m(s)} \pi_j(A_j | \mathbf{H}_j)},$$

the Radon–Nikodym derivative $d\mathcal{P}^{\tilde{\mathcal{D}}_{m(s)}}/d\mathcal{P}_{m(s)}$ exists under the positivity assumption that $\pi_j(A_j | \mathbf{H}_j) > 0$ for all $1 \leq j \leq m(s)$. Thus, the expectation $E^{\tilde{\mathcal{D}}_{m(s)}}$ in (7) can be expressed as an unconditional one:

$$E^{\tilde{\mathcal{D}}_{m(s)}}[I(T \geq s)] = E \left\{ \frac{I[\tilde{\mathbf{A}}_{m(s)} = \tilde{\mathcal{D}}_{m(s)}(\mathbf{H}_{m(s)})]}{\prod_{j=1}^{m(s)} \pi_j(A_j | \mathbf{H}_j)} I(T \geq s) \right\}, \quad (8)$$

which incorporates the inverse weighting of the treatment probability $\prod_{j=1}^{m(s)} \pi_j(A_j | \mathbf{H}_j)$. This probability appears in randomized clinical trials or must be estimated in observational studies. Similarly, we have

$$E^{\tilde{\mathcal{D}}_{m(s)}}[I(T = s)] = E \left\{ \frac{I[\tilde{\mathbf{A}}_{m(s)} = \tilde{\mathcal{D}}_{m(s)}(\mathbf{H}_{m(s)})]}{\prod_{j=1}^{m(s)} \pi_j(A_j | \mathbf{H}_j)} I(T = s) \right\}. \quad (9)$$

However, the survival time T in (8) and (9) is not fully observed due to censoring. In practice, only the $Y = T \wedge C$ is observed. Note that we have assumed that the censoring time C is independent of $(T, \tilde{\mathbf{A}}_m, \tilde{\mathbf{X}}_m)$ in [Assumption 1](#) in [Section 2](#). In fact, this assumption is standard in clinical trials with several follow-up studies ([Jiang et al. 2017a](#)). Under this noninformative censoring assumption, we can substitute the unobserved survival time T with the observed Y by

$$\begin{aligned} &1 - q^{\tilde{\mathcal{D}}_{m_g}}(s) \\ &= \frac{E^{\tilde{\mathcal{D}}_{m(s)}}[I(T = s)] E[I(C \geq s)]}{E^{\tilde{\mathcal{D}}_{m(s)}}[I(T \geq s)] E[I(C \geq s)]} \\ &= \frac{E\{I[\tilde{\mathbf{A}}_{m(s)} = \tilde{\mathcal{D}}_{m(s)}(\mathbf{H}_{m(s)})] I(Y = s) \Delta / \prod_{j=1}^{m(s)} \pi_j(A_j | \mathbf{H}_j)\}}{E\{I[\tilde{\mathbf{A}}_{m(s)} = \tilde{\mathcal{D}}_{m(s)}(\mathbf{H}_{m(s)})] I(Y \geq s) / \prod_{j=1}^{m(s)} \pi_j(A_j | \mathbf{H}_j)\}}. \end{aligned}$$

Consequently, we can estimate the survival probability $S^{\tilde{\mathcal{D}}_{m_g}}(g)$ with the observed data. That is,

$$\bar{S}^{\tilde{\mathcal{D}}_{m_g}}(g) = \prod_{s=1}^g \bar{q}^{\tilde{\mathcal{D}}_{m_g}}(s) = \prod_{s=1}^g \left\{ 1 - \frac{\sum_{i=1}^n \bar{w}_i(s) I(Y_i = s) \Delta_i}{\sum_{i=1}^n \bar{w}_i(s) I(Y_i \geq s)} \right\}, \quad (10)$$

where

$$\bar{w}_i(s) = \frac{I[\tilde{\mathbf{A}}_{im(s)} = \tilde{\mathcal{D}}_{m(s)}(\mathbf{H}_{im(s)})]}{\prod_{j=1}^{m(s)} \hat{\pi}_j(A_{ij} | \mathbf{H}_{ij})}, \quad (11)$$

and $\hat{\pi}_j(A_{ij} | \mathbf{H}_{ij})$ is an estimator of $\pi_j(A_{ij} | \mathbf{H}_{ij})$ for $1 \leq j \leq m_g$, which depends on the number of possible treatment choices and can be estimated through a penalized multinomial model in [Section 5](#).

One crucial component of the proposed method is decomposing the survival function in terms of survival probabilities at time-points before the given g as in (10). This strategy allows us to only consider treatments received at stages before these time-points. Thus, we can incorporate all available information from patients who are subject to censoring even before the given g . In addition, our method integrates the estimations of decision rules at multiple stages together instead of recursively fitting the regression model at each stage, which is more robust to model-misspecification.

3.2. Multicategory Dynamic Treatment Regimes

Although many developments have been made in estimation of the optimal DTR for binary treatments, few existing approaches can handle multicategory treatments. In fact, the extension from binary treatment choices to multicategory treatments is nontrivial. For example, one can use a sequential binary classifier such as the one-versus-rest approach, but this could lead to inconsistent estimation ([Zhang et al. 2020](#)). In addition, it is difficult to make multicategory treatment comparisons efficiently ([Tao and Wang 2017](#)).

To overcome these difficulties, we propose an angle-based weighted KM method to obtain the optimal DTR $\tilde{\mathcal{D}}_{m_g}^*$, such that $S^{\tilde{\mathcal{D}}_{m_g}^*}(g) \geq S^{\tilde{\mathcal{D}}_{m_g}}(g)$ for all sequential rules $\tilde{\mathcal{D}}_{m_g}$ under the multicategory treatment framework. We adopt an angle-based classification ([Zhang and Liu 2014](#)) idea to incorporate multicategory treatments. Under the multicategory DTR framework, there are K possible treatments. To visualize the K possible treatments from a geometric perspective, we define a simplex $\mathbf{V}$ with K vertices $(\mathbf{V}_1, \dots, \mathbf{V}_j, \dots, \mathbf{V}_K)$ in a $(K-1)$ -dimensional space such that

$$\mathbf{V}_j = \begin{cases} \frac{1}{\sqrt{K-1}} \mathbf{1}_{K-1}, & j = 1, \\ -\frac{1+\sqrt{K}}{(K-1)^{3/2}} \mathbf{1}_{K-1} + \sqrt{\frac{K}{K-1}} \mathbf{e}_{j-1}, & 2 \leq j \leq K, \end{cases}$$

where $\mathbf{1}_{K-1}$ is a vector of 1's with a length of $K-1$, and $\mathbf{e}_j \in \mathbb{R}^{K-1}$ is a vector with 1 in the j th element and 0 elsewhere. Here $\mathbf{V}_j$ represents a potential treatment choice. Since this simplex is symmetric for all vertices, the angle between any pair of vertices $\mathbf{V}_i$ and $\mathbf{V}_j$ is the same for $1 \leq i, j \leq K$.

Under this geometric framework, we use the angles between a $(K-1)$ -dimensional classification function vector $\mathbf{f}_m(\mathbf{H}_m) =$

$(f_{m,1}(\mathbf{H}_m), \dots, f_{m,K-1}(\mathbf{H}_m))^T$ and $V_1, V_2, \dots, V_K$ to determine the decision rule at the m th stage for $1 \leq m \leq m_g$. This decision rule follows a principle which identifies the smallest angle. Specifically, the decision rule at the m th stage is defined as

$$\begin{aligned} \mathcal{D}_m(\mathbf{H}_m) &= \operatorname{argmin}_{A \in \{1, \dots, K\}} \operatorname{Ang}(V_A, f_m(\mathbf{H}_m)) \\ &= \operatorname{argmax}_{A \in \{1, \dots, K\}} \langle V_A, f_m(\mathbf{H}_m) \rangle, \end{aligned} \quad (12)$$

where $\operatorname{Ang}(V_A, f_m(\mathbf{H}_m))$ represents the angle between V_A and $f_m(\mathbf{H}_m)$. The last equality in (12) follows from the fact that minimizing the angle between V_A and $f_m(\mathbf{H}_m)$ is equivalent to maximizing the inner product between V_A and $f_m(\mathbf{H}_m)$. We let $K = 3$ as an illustration example in Figure 1, where V_1 ,

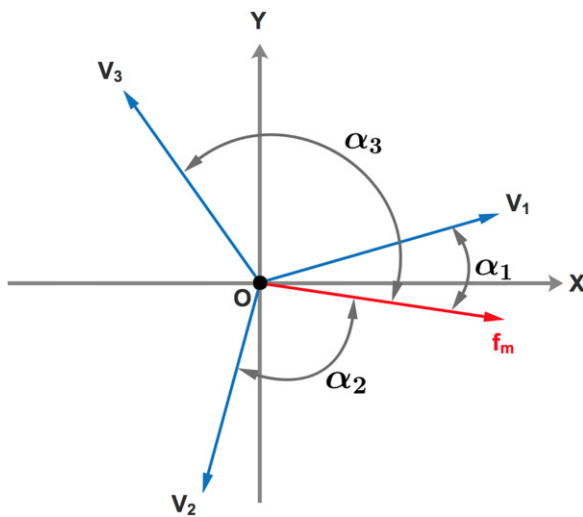


Figure 1. Illustration of the angle-based approach with $K = 3$.

V_2 , and V_3 represent the three possible treatment choices. Then the decision corresponding to the classification function f_m in Figure 1 is treatment 1, since the angle α_1 between f_m and V_1 is the smallest among α_1, α_2 , and α_3 .

Substituting (12) for (11), then maximizing the function in (10) becomes a maximization problem of $\tilde{S}^{\mathcal{D}_{m_g}}(g)$ over f_m . However, it is still challenging to maximize $\tilde{S}^{\mathcal{D}_{m_g}}(g)$ directly, since $\tilde{S}^{\mathcal{D}_{m_g}}(g)$ is discontinuous with respect to f_m for $1 \leq m \leq m_g$ due to the discontinuity of the indicator function in (11). Alternatively, a computationally more efficient approach is to replace the indicator by a continuous surrogate function. We consider adopting a logistic function

$$l(u) = \frac{1}{1 + e^{-b(u-u_0)}}, \quad (13)$$

which is a smooth and strictly increasing function, where the coefficient $b > 0$ measures the steepness of the logistic curve and $u_0 \in \mathbb{R}$ represents the inflection point. We illustrate the logistic function with various b and u_0 in Figure 2. The choices of the b and u_0 will be discussed in Section 5.

We propose to replace $I[A_{im} = \mathcal{D}_m(\mathbf{H}_{im})]$ with $l(\langle V_{A_{im}}, f_{im}(\mathbf{H}_{im}) \rangle)$ for $1 \leq i \leq n$ and $1 \leq m \leq m_g$ in (11). That is, the proposed estimator $\hat{F}$ is the maximizer of the following penalized log-transformed survival function

$$Q_{n,g}(F) = \log \left\{ \tilde{S}^{\mathcal{D}_{m_g}}(g) \right\} - J_\lambda(F), \quad (14)$$

where

$$\begin{aligned} \tilde{S}^{\mathcal{D}_{m_g}}(g) &= \prod_{s=1}^g \hat{q}^{\mathcal{D}_{m_g}}(s) \\ &= \prod_{s=1}^g \left\{ 1 - \frac{\sum_{i=1}^n w_i(s) I(Y_i = s) \Delta_i}{\sum_{i=1}^n w_i(s) I(Y_i \geq s)} \right\}, \end{aligned} \quad (15)$$

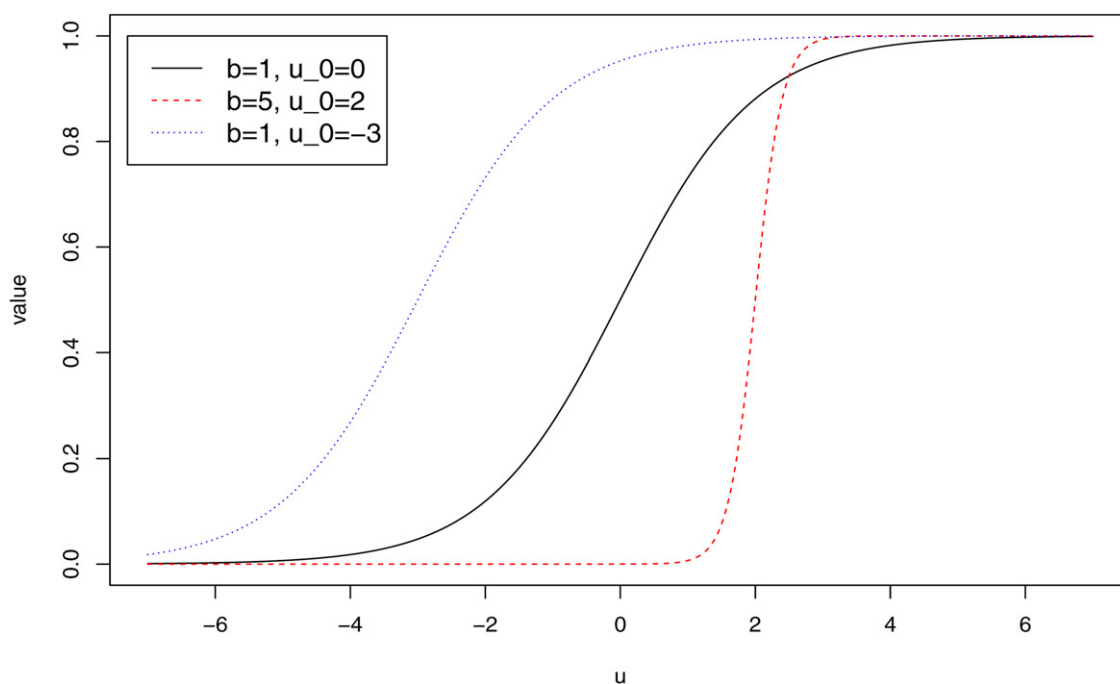


Figure 2. Illustration of the logistic function $l(u)$ with different b and u_0 .

$$w_i(s) = \frac{\prod_{j=1}^{m(s)} l\left(\left\langle \mathbf{V}_{A_{ij}}, \mathbf{f}_j(\mathbf{H}_{ij}) \right\rangle\right)}{\prod_{j=1}^{m(s)} \pi_j(A_{ij}|\mathbf{H}_{ij})}, \quad (16)$$

$\mathbf{F} = (\mathbf{f}_1, \dots, \mathbf{f}_{m_g})$, and $J_\lambda(\mathbf{F})$ is a penalty function on $\mathbf{F}$ with tuning parameter λ for preventing overfitting and avoiding the identifiability issue.

4. Theory

4.1. Fisher Consistency

In this subsection, we demonstrate the Fisher consistency (Liu 2007) of the proposed method, indicating that the proposed estimator achieves the optimal DTR if the estimator is calculated using the entire population. Although the entire population is typically not observed in practice, the Fisher consistency shows that the proposed method is able to predict the best treatment for each subject if we have sufficient information. Specifically, we show that the DTR corresponding to the proposed estimator $\bar{\mathbf{F}}$ based on the entire population recommends the same treatments as the optimal $\bar{\mathbf{D}}_{m_g}^*$, where $\bar{\mathbf{D}}_{m_g}^*$ is the DTR that maximizes the true survival function $S^{\bar{\mathbf{D}}_{m_g}^*}(g)$ and $\bar{\mathbf{F}} = (\bar{\mathbf{f}}_1, \dots, \bar{\mathbf{f}}_{m_g})$ is the maximizer of

$$R_g(\mathbf{F}) = \sum_{s=1}^g \log\{\tilde{q}(s, \mathbf{F})\} \quad (17)$$

over $\mathcal{F} = \left\{ \mathbf{F} : \mathbf{f}_j \text{ is measurable for } 1 \leq j \leq m_g \text{ and } \|\mathbf{F}\|^2 \leq \mu^* \right.$
a.s. $\left. \right\}$. Here, $\mu^* = \|\mathbf{F}^*\|^2$ and

$$\begin{aligned} \tilde{q}(s, \mathbf{F}) & \quad (18) \\ &= 1 - \frac{E\left[\prod_{j=1}^{m(s)} l\left(\left\langle \mathbf{V}_{A_j}, \mathbf{f}_j(\mathbf{H}_j) \right\rangle\right) I(Y=s) \Delta / \left\{ \prod_{j=1}^{m(s)} \pi_j(A_j|\mathbf{H}_j) \right\}\right]}{E\left[\prod_{j=1}^{m(s)} l\left(\left\langle \mathbf{V}_{A_j}, \mathbf{f}_j(\mathbf{H}_j) \right\rangle\right) I(Y \geq s) / \left\{ \prod_{j=1}^{m(s)} \pi_j(A_j|\mathbf{H}_j) \right\}\right]}, \end{aligned}$$

where $\mathbf{F}^*$ consists of classification functions corresponding to $\bar{\mathbf{D}}_{m_g}^*$, $\|\mathbf{F}\|^2 = \sum_{j=1}^{m_g} \sum_{i=1}^{K-1} \|\mathbf{f}_{ji}\|_2^2$, and $\|\mathbf{f}_{ji}\|_2 = \sup_{\mathbf{H}_j} |\mathbf{f}_{ji}(\mathbf{H}_j)| / \|\mathbf{H}_j\|_2$. When $\mathbf{f}_{ji}(\mathbf{H}_j) = \mathbf{H}_j^T \boldsymbol{\theta}_{ji}$ is a linear function, $\|\mathbf{f}_{ji}\|_2 = \|\boldsymbol{\theta}_{ji}\|_2$ is consistent with the L_2 penalty in Section 5, where $\boldsymbol{\theta}_{ji}$ is a parameter vector with the same dimension as $\mathbf{H}_j$.

Condition 1. Suppose that each covariate and treatment A_j are bounded by a constant C_x with probability 1. The $l(\cdot)$ in (18) is a strictly increasing surrogate function such that $l(0) > \bar{c}$ for some constant $\bar{c} \in (0, 1/2]$, $l(u) > 0$, and the second derivative $l''(u) < 0$ for $|u| \leq C_H \sqrt{\mu^*}$, where $C_H = C_x \sqrt{pm_g + m_g - 1}$.

Condition 1 requires that each covariate is bounded, which is satisfied in general practice and is also assumed in Zhang and Liu (2014). The requirements for $l(\cdot)$ in Condition 1 hold when the surrogate function $l(\cdot)$ is positive, strictly increasing, and concave in a given region, for example, the logistic function in (13) with $u_0 \leq -C_H \sqrt{\mu^*}$.

To simplify expression of the following conditions and theorems, we first define some notations. Let $\tilde{\mathbf{A}}_{w^c}$ be a vector

consisting of A_j for $1 \leq j \leq w-1$ and $w+1 \leq j \leq m_g$. For $1 \leq s \leq G$ and $1 \leq w, m \leq m_g$, we let

$$\begin{aligned} \varphi(s, \tilde{\mathbf{x}}_w, a_w, \tilde{\mathbf{a}}_{w^c}) &= E_{w+1}^{\tilde{\mathbf{A}}_w} \left[\dots E_{m_g}^{\tilde{\mathbf{A}}_{m_g-1}} \left\{ P(T \geq s | \tilde{\mathbf{X}}_w = \tilde{\mathbf{x}}_w, A_w = a_w, \tilde{\mathbf{A}}_{w^c} = \tilde{\mathbf{a}}_{w^c}) \right\} \dots \right], \\ \eta_1(\tilde{\mathbf{X}}_w, \tilde{\mathbf{A}}_{w^c}, s, m) &= \min_{a_w \neq D_w^*(\mathbf{H}_w)} \left\{ c_L^{m-w} \varphi(s, \tilde{\mathbf{X}}_w, D_w^*(\mathbf{H}_w), \tilde{\mathbf{A}}_{w^c}) \right. \\ &\quad \left. - c_U^{m-w} \varphi(s, \tilde{\mathbf{X}}_w, a_w, \tilde{\mathbf{A}}_{w^c}) \right\}, \\ \eta_2(\tilde{\mathbf{X}}_w, \tilde{\mathbf{A}}_{w^c}, s, m) &= \max_{a_w \neq D_w^*(\mathbf{H}_w)} \left\{ \frac{c_U^{m-w} c'_U}{c'_L} \varphi(s, \tilde{\mathbf{X}}_w, D_w^*(\mathbf{H}_w), \tilde{\mathbf{A}}_{w^c}) \right. \\ &\quad \left. - c_L^{m-w} \varphi(s, \tilde{\mathbf{X}}_w, a_w, \tilde{\mathbf{A}}_{w^c}) \right\}, \end{aligned}$$

where $E_{m-1}^{\tilde{\mathbf{A}}_{m-1}}$ represents an expectation with respect to $\{\mathbf{X}_m | \tilde{\mathbf{X}}_{m-1}, \tilde{\mathbf{A}}_{m-1}\}$ for $w+1 \leq m \leq m_g$, c'_L and c'_U are the lower and upper bounds of the first derivative $l'(u)$, respectively, c_L and c_U are the lower and upper bounds of $l(u)$, respectively, for $|u| \leq C_H \sqrt{\mu^*}$.

Condition 2. For each stage $1 \leq w \leq m_g$, we assume

$$\begin{aligned} \eta_1(\tilde{\mathbf{X}}_w, \tilde{\mathbf{A}}_{w^c}, g, m_g) &- \sum_{i=w}^{m_g-1} \eta_1^-(\tilde{\mathbf{X}}_w, \tilde{\mathbf{A}}_{w^c}, v(i+1), i) \\ &- \sum_{i=w}^{m_g} \eta_2^+(\tilde{\mathbf{X}}_w, \tilde{\mathbf{A}}_{w^c}, v(i), i) / \kappa_i > 0, \quad (19) \end{aligned}$$

where η^+ and η^- represent the positive and negative parts of η , respectively, κ_i is the lower bound of $\prod_{j=i+1}^{m_g} \pi_j(A_j|\mathbf{H}_j)$ for $1 \leq i \leq m_g - 1$ with $\kappa_{m_g} = 1$, and $v(m)$ is the first time-point at the m th stage for $1 \leq m \leq m_g$. If $w = m_g$, then the second term in (19) vanishes.

Condition 2 ensures that receiving the optimal individualized treatment at each stage increases the targeted survival probability at the given time g . It also ensures that the increase in the survival probability at the given time g exceeds the changes of survival probabilities at the beginning of stages $1 \leq m \leq m_g$.

Theorem 1. Under Assumptions 1–3 and Conditions 1 and 2, we have

$$\operatorname{argmax}_{A \in \{1, \dots, K\}} \left\langle \mathbf{V}_A, \bar{\mathbf{f}}_j(\mathbf{H}_j) \right\rangle = D_j^*(\mathbf{H}_j) \text{ almost surely,}$$

for stages $j = 1, \dots, m_g$.

Theorem 1 states that the proposed method is Fisher consistent, which is a fundamental property for a classifier. **Theorem 1** shows that the proposed method recommends the optimal treatment for each subject at each stage with probability 1 if we obtain sufficient data. To the best of our knowledge, this is the first Fisher consistency result for DTR under the angle-based multicategory framework.

4.2. Risk Bound

In this subsection, we establish the excess risk bound theory on solving the survival probability maximization problem. We particularly focus on a linear classification function framework, that is, $f_{ji}(\mathbf{H}_j) = \mathbf{H}_j^T \boldsymbol{\theta}_{ji}$ for $1 \leq j \leq m_g$, $1 \leq i \leq K-1$, and $J_\lambda(\mathbf{F}) = \lambda \sum_{j=1}^{m_g} \sum_{i=1}^{K-1} \|\boldsymbol{\theta}_{ji}\|_2^2$. Define

$$\mathcal{F}(\mu) = \left\{ \mathbf{F} : f_{ji}(\mathbf{H}_j) = \mathbf{H}_j^T \boldsymbol{\theta}_{ji} \text{ for } 1 \leq j \leq m_g \text{ and } 1 \leq i \leq K-1, \sum_{j=1}^{m_g} \sum_{i=1}^{K-1} \|\boldsymbol{\theta}_{ji}\|_2^2 \leq \mu \right\},$$

and $\widehat{\mathbf{F}}_n$ be the maximizer of (14) under the linear classification function framework. Then, $\widehat{\mathbf{F}}_n$ is also the maximizer of

$$R_{n,g}(\mathbf{F}) = \log \left\{ \widetilde{S}^{\mathcal{D}_{m_g}}(g) \right\} \quad (20)$$

over $\mathcal{F}(\mu_n)$ with $\mu_n = \|\widehat{\mathbf{F}}_n\|^2$. We refer to $\mathbf{F}_l = \arg\max_{\mathbf{F} \in \mathcal{F}(\mu_n)} R_g(\mathbf{F})$ as the best classification function within $\mathcal{F}(\mu_n)$.

Following notations in Zhang and Liu (2014) and Zhao et al. (2015), we define $d_l(\widehat{\mathbf{F}}_n, \mathbf{F}_l) = R_g(\mathbf{F}_l) - R_g(\widehat{\mathbf{F}}_n)$ and $d(\widehat{\mathbf{F}}_n, \mathbf{F}_l) = R_g^*(\mathbf{F}_l) - R_g^*(\widehat{\mathbf{F}}_n)$ as the excess l -risk and the excess risk for $\widehat{\mathbf{F}}_n$, respectively, where

$$R_g^*(\mathbf{F}) = \log \{ S^{\mathcal{D}_{m_g}}(g) \} \quad (21)$$

is the true log-survival function. The excess l -risk $d_l(\widehat{\mathbf{F}}_n, \mathbf{F}_l)$ measures difference between the proposed estimator $\widehat{\mathbf{F}}_n$ and the best classification function $\mathbf{F}_l$ in terms of the R_g in (17), while $d(\widehat{\mathbf{F}}_n, \mathbf{F}_l)$ measures difference between $\widehat{\mathbf{F}}_n$ and $\mathbf{F}_l$ in terms of the true log-survival function. The following theorems provides bounds of the excess l -risk and the excess risk for large sample size.

Condition 3. The $l(\cdot)$ is a positive and bounded Lipschitz function. We assume each covariate is bounded with probability one.

Similarly to Condition 1, Condition 3 also requires the bounds of the surrogate function $l(\cdot)$ and covariates. The Lipschitz assumption in Condition 3 ensures that the first derivative of $l(\cdot)$ is bounded, which is also assumed in Zhang and Liu (2014). Indeed, the logistic function in (13) satisfies Condition 3 regarding $l(\cdot)$.

Theorem 2. Under Assumptions 1–3 and Condition 3, for any constant $c_1 > 0$, there exists a constant $c_2 > 0$ such that

$$0 < d_l(\widehat{\mathbf{F}}_n, \mathbf{F}_l) \leq c_1 n^{-\alpha}$$

with probability at least $1 - \exp(-c_2 n^{1-2\alpha})$ for sufficiently large n , where $\alpha \in (0, 1/2)$ and c_2 may depend on c_1, g, m_g, p and lower bounds of $\pi_j (1 \leq j \leq m_g)$.

Theorem 2 establishes that the excess l -risk converges to 0 as $n \rightarrow \infty$ with the convergence rate of nearly $n^{-1/2}$ and the probability exponentially tending to 1.

Condition 4. There exist constants $\tau_1 \geq 1$ and $\tau_2, \tau_3, \tau_4 > 0$ such that for all small $\epsilon > 0$,

$$\inf_{\{\mathbf{F} \in \mathcal{F}(\mu_n) : \rho(\mathbf{F}, \mathbf{F}_l) \geq \epsilon\}} d_l(\mathbf{F}, \mathbf{F}_l) \geq \tau_2 \epsilon^{\tau_1}, \quad (22)$$

$$\sup_{\{\mathbf{F} \in \mathcal{F}(\mu_n) : \rho(\mathbf{F}, \mathbf{F}_l) \leq \epsilon\}} |d(\mathbf{F}, \mathbf{F}_l)| \leq \tau_3 \epsilon^{\tau_4}, \quad (23)$$

where $\rho(\mathbf{F}, \mathbf{F}_l) = \|\mathbf{F} - \mathbf{F}_l\|$.

Condition 4 controls the excess l -risk and the excess risk in a neighborhood of $\mathbf{F}_l$ via the norm of $\mathbf{F}$, which is also imposed in Zhang and Liu (2014). We provide scenarios where Condition 4 holds in Proposition 1 in the supplementary materials. Under Condition 4, we establish the excess risk based on the true log-survival function in the following theorem.

Theorem 3. Under Assumptions 1–3 and Conditions 3 and 4, for any constant $c_3 > 0$, there exists a constant $c_4 > 0$ such that

$$|d(\widehat{\mathbf{F}}_n, \mathbf{F}_l)| \leq c_3 n^{-\tau_4 \alpha / \tau_1}$$

with probability at least $1 - \exp(-c_4 n^{1-2\alpha})$ for sufficiently large n , where $\alpha \in (0, 1/2)$ and c_4 depends on c_3, g, m_g, p and lower bounds of $\pi_j (1 \leq j \leq m_g)$.

5. Implementation

In this section, we demonstrate how to solve the maximization problem with the objective function $Q_{n,g}(\boldsymbol{\theta})$ in (14) and how to select proper tuning parameters. Specifically, we consider that the classification functions follow the parametric model $\mathbf{F}(\mathbf{H}_{m_g}) = \mathbf{F}(\mathbf{H}_{m_g}; \boldsymbol{\theta})$. An estimator $\widehat{\boldsymbol{\theta}}$ is obtained through minimizing $-Q_{n,g}(\boldsymbol{\theta})$ with a L_2 penalty, that is,

$$\begin{aligned} \widehat{\boldsymbol{\theta}} &= \arg \min_{\boldsymbol{\theta}} -Q_{n,g}(\boldsymbol{\theta}) \\ &= \arg \min_{\boldsymbol{\theta}} \left[-\sum_{s=1}^g \log \left\{ 1 - \frac{\sum_{i=1}^n w_i(s) I(Y_i = s) \Delta_i}{\sum_{i=1}^n w_i(s) I(Y_i \geq s)} \right\} \right. \\ &\quad \left. + \lambda \|\boldsymbol{\theta}\|_2^2 \right], \end{aligned} \quad (24)$$

where

$$w_i(s) = \frac{\prod_{j=1}^{m(s)} l(\langle \mathbf{V}_{A_{ij}}, \mathbf{f}_j(\mathbf{H}_{ij}; \boldsymbol{\theta}) \rangle)}{\prod_{j=1}^{m(s)} \pi_j(A_{ij} | \mathbf{H}_{ij})}, \quad (25)$$

$\mathbf{f}_j(\mathbf{H}_{ij}; \boldsymbol{\theta}) = (\mathbf{H}_{ij}^T \boldsymbol{\theta}_{j1}, \dots, \mathbf{H}_{ij}^T \boldsymbol{\theta}_{jK-1})^T$, and $\boldsymbol{\theta}$ is a vector consisting of all $\boldsymbol{\theta}_{jk}$ for $1 \leq j \leq m_g$ and $1 \leq k \leq K-1$. The objective function $-Q_{n,g}(\boldsymbol{\theta})$ in (24) is a nonconvex but smooth function. We adopt the Broyden–Fletcher–Goldfarb–Shanno (BFGS, Fletcher 1987) algorithm to solve (24). The BFGS algorithm converges since it incorporates a line search for a proper step size to ensure a decrease of the objective function at each step. Note that the BFGS algorithm can only converge to a local minimizer due to the nonconvexity of $-Q_{n,g}(\boldsymbol{\theta})$. However, a well-chosen initialization can lead to a good solution (Geem 2006; Diamond, Takapoui, and Boyd 2018). For implementation, we try different initial values and select the one with the smallest value of the objective function. The details of the algorithm are summarized in Algorithm 1.

Algorithm 1 BFGS algorithm

- 1: Input all observed $\{(Y_i, \Delta_i, \mathbf{X}_{it}, A_{it}) : t = 1, \dots, m_g\}_{i=1}^n$, an initial step size η_0 , an initial value θ_0 , an approximate Hessian matrix $\mathbf{B}_0 = \mathbf{I}$ and a stopping criterion $\varepsilon = 10^{-4}$.
- 2: At the $(k+1)$ th iteration ($k \geq 0$), estimate θ_{k+1} as follows

$$\hat{\theta}_{k+1} = \hat{\theta}_k + \eta_k \mathbf{B}_k^{-1} \nabla Q_{n,g}(\hat{\theta}_k),$$

where $\nabla Q_{n,g}(\hat{\theta}_k)$ is the gradient value of $Q_{n,g}(\theta)$ at $\hat{\theta}_k$, and the step size η_k is chosen by the line search method in the direction found in the first step, so $\eta_k = \arg \min -Q_{n,g}(\hat{\theta}_k + \eta_k \mathbf{B}_k^{-1} \nabla Q_{n,g}(\hat{\theta}_k))$.

- 3: Calculate $\mathbf{s}_k = \hat{\theta}_{k+1} - \hat{\theta}_k$ and $\mathbf{z}_k = \nabla Q_{n,g}(\hat{\theta}_k) - \nabla Q_{n,g}(\hat{\theta}_{k+1})$, and then obtain the approximate inverse of the matrix $\mathbf{B}_{k+1}$ as follows

$$\mathbf{B}_{k+1}^{-1} = \mathbf{B}_k^{-1} + \frac{(\mathbf{s}_k^\top \mathbf{z}_k + \mathbf{z}_k^\top \mathbf{B}_k^{-1} \mathbf{z}_k)(\mathbf{s}_k \mathbf{s}_k^\top)}{(\mathbf{s}_k^\top \mathbf{z}_k)^2} - \frac{\mathbf{B}_k^{-1} \mathbf{z}_k \mathbf{s}_k^\top + \mathbf{s}_k \mathbf{z}_k^\top \mathbf{B}_k^{-1}}{\mathbf{s}_k^\top \mathbf{z}_k}.$$

- 4: Stop if $\max\{\|\mathbf{s}_k\|^2, \|\mathbf{z}_k\|^2\} < \varepsilon$. Otherwise go to Steps 2 and 3.

In practice, a proper choice of the parameters u_0 , b in (13) and λ in (14) is critical for estimation of the optimal DTR. In particular, the choice of the tuning parameter λ controls model complexity. We let $u_0 = -C_H \sqrt{C_Q/\lambda}$ to satisfy Condition 1 for any given b and λ , where

$$C_Q = -\sum_{s=1}^g \log \left[1 - \frac{n^{-1} \sum_{i=1}^n I(Y_i = s) \Delta_i / \left\{ \prod_{j=1}^{m(s)} \pi_j(A_{ij} | \mathbf{H}_{ij}) \right\}}{n^{-1} \sum_{i=1}^n I(Y_i \geq s) / \left\{ \prod_{j=1}^{m(s)} \pi_j(A_{ij} | \mathbf{H}_{ij}) \right\}} \right]. \quad (26)$$

Various tuning techniques have been put forward in the existing literature. Here, we briefly discuss the d -fold cross-validation procedure, which is commonly used in practice. Specifically, all the samples are randomly partitioned into d equally sized subsets. For each pair of tuning parameters (b, λ) , we fit the model based on every $d-1$ subsets, and calculate the estimated survival function in (10) at the given time g for the remaining subset. Equivalently, the $d-1$ subsets form a training set, while the remaining subset serves as a validation set. We obtain a total of d training sets and d validation sets, and choose the optimal tuning parameters which maximize the average of the estimated survival probabilities in (10) over all of the validation sets. That is,

$$(\hat{b}, \hat{\lambda})_{\text{validated}} = \operatorname{argmax}_{(b, \lambda)} \frac{1}{d} \sum_{r=1}^d \tilde{\mathcal{S}}_r^{m_g}(g|b, \lambda),$$

where $\tilde{\mathcal{D}}_{m_g}$ depends on the proposed estimator $\hat{\theta}$ given b and λ using the r th training set, and $\tilde{\mathcal{S}}_r^{m_g}(g|b, \lambda)$ is the estimated survival probability at the given time g based on $\hat{\theta}$ and the corresponding validation set.

For data from an observational study, at the m th stage ($1 \leq m \leq m_g$), the propensity score $\pi_m(A_{im} | \mathbf{H}_{im})$ is unknown and thus need to be estimated. In this article, we use a L_1 -penalized multinomial logistic regression model (Friedman, Hastie, and Tibshirani 2010) to estimate the propensity scores. We adopt a

symmetric parameterization for this model, that is, for each $1 \leq k \leq K$,

$$P(A_m = k | \mathbf{H}_m) = \frac{\exp(\beta_{0k} + \boldsymbol{\beta}_k^\top \mathbf{H}_m)}{\sum_{j=1}^K \exp(\beta_{0j} + \boldsymbol{\beta}_j^\top \mathbf{H}_m)},$$

where β_{0k} is an intercept and $\boldsymbol{\beta}_k$ consists of parameters for $\mathbf{H}_m$. We can minimize the corresponding negative log-likelihood function:

$$\begin{aligned} & -\log \{L(\beta_{01}, \boldsymbol{\beta}_1, \dots, \beta_{0K}, \boldsymbol{\beta}_K)\} \\ &= -\sum_{i=1}^n \left[\sum_{k=1}^K I(A_{im} = k) (\beta_{0k} + \boldsymbol{\beta}_k^\top \mathbf{H}_{im}) \right. \\ & \quad \left. - \log \left\{ \sum_{j=1}^K \exp(\beta_{0j} + \boldsymbol{\beta}_j^\top \mathbf{H}_{im}) \right\} \right] \end{aligned}$$

to obtain parameter estimation. However, there exists an over-parameterization issue without constraints on the parameters, since different values of the parameters $\{\beta_{0k}, \boldsymbol{\beta}_k\}_{k=1}^K$ and $\{\beta_{0k} - c, \boldsymbol{\beta}_k - \mathbf{c}\}_{k=1}^K$ may give an identical probability $P(A_m = k | \mathbf{H}_m)$. Therefore, we add L_1 -penalty to the negative log-likelihood function as in Friedman, Hastie, and Tibshirani (2010). We estimate β_{0k} and $\boldsymbol{\beta}_k$ ($1 \leq k \leq K$) through minimizing the following penalized negative log-likelihood function

$$-\log \{L(\beta_{01}, \boldsymbol{\beta}_1, \dots, \beta_{0K}, \boldsymbol{\beta}_K)\} + \lambda^* \sum_{k=1}^K \|(\beta_{0k}, \boldsymbol{\beta}_k)\|_1,$$

where $\|\cdot\|_1$ denotes the L_1 norm, and λ^* is a tuning parameter and can be tuned by cross-validation. If the assumption of the multinomial logistic regression model is violated, an alternative approach is to apply nonparametric methods (Cham 2013; McCaffrey et al. 2013; Pan and Bai 2015) to estimate the propensity score.

6. Simulation Studies

For numerical experiments, we construct five simulation examples including multicategory settings and binary treatments settings to evaluate the finite sample performance of the proposed method. We present Examples 1–3 in this section, and provide Examples 4 and 5 in the supplementary materials. Examples 1, 2, 4, and 5 are multicategory settings, where we compare the proposed method with the censored Q-learning method (Goldberg and Kosorok 2012) and the subgroup identification method (subgroup) (Huling and Yu 2018). In Example 3, we restrict the proposed method to a binary treatment setting which is a special case of our multicategory framework. Besides the above competing methods, we also compare the proposed method with a dynamic weighted survival modeling (DWSurv) method (Simoneau et al. 2020) in Example 3, which is proposed to estimate binary optimal dynamic treatments for censored data. To evaluate the performance of each method, we use the estimated conditional survival probability in (10) under the recommended DTR as a criterion.

The subgroup method can only handle binary treatments, we extend it via a one-versus-rest strategy to handle multicategory

treatments for fair comparisons. For each treatment category k ($1 \leq k \leq K$), we train a binary classifier by labeling the samples of the treatment category k as positive and others as negative. We choose the treatment category which has the largest value of classification function as the recommended treatment category.

In addition, the subgroup method is designed to estimate the optimal decision rule at a single stage. To extend it to handle the multiple stages, we adjust the observed survival time following Goldberg and Kosorok (2012) and Zhao et al. (2018). Specifically, we first calculate within stage survival time for each stage. The within stage survival time for the i th stage is defined as the difference between the observed survival time Y and the beginning time point at the $m(Y)$ th stage if $i = m(Y)$, or the length of the i th stage if $i < m(Y)$. At each stage, we fit the subgroup identification model by treating the within stage survival time as an outcome. The recommended treatment is determined based on the fitted subgroup model. Then, we combine the recommended treatment at each stage, and obtain a sequential treatment recommendation for all stages.

In our proposed method, the tuning parameter λ is tuned from a candidate tuning set $(0.0001, 0.0005, 0.001, 0.005, 0.01, 0.05, 0.1)$, and the parameter b takes values from 0.5 to 10 with a step of 0.5. The range of the regularized parameter λ^* in the propensity score estimation is chosen from 0.0001 to 1 with a step of 0.0001. All tuning parameters are selected by 5-fold cross-validation. For each λ , we calculate $u_0 = -C_H \sqrt{C_Q/\lambda}$, where C_H and C_Q are defined in Condition 1 and Equation (26), respectively.

We generate data $(Y_i, C_i, T_i, \Delta_i, \mathbf{X}_{im}, A_{im})$ for $i = 1, \dots, n$ and $m = 1, \dots, M$, where M is the number of total stages, $\mathbf{X}_{im}$ consists of p covariates for the i th patient at the m th stage, and A_{im} is the treatment assigned to the i th patient at the m th stage. We let $p = 25$ and $M = 5$ in all examples. We consider three potential treatment choices $\{1, 2, 3\}$ for Examples 1, 2, 4, and 5, and binary treatment choices $\{1, -1\}$ for Example 3. We generate a training set with 500 observations for model fitting, and simulate an independent testing set with 2000 observations for evaluation of model performance. We run simulations 150 times and report the average values of estimated conditional survival probabilities. We consider three different censoring rates and provide boxplots of the estimated conditional survival probabilities. Let d_{im} be the ideal treatment choice for the i th subject at the m th stage. In the following examples, the decision time-points are 0.5, 1, 1.5, 2, and 2.5. We specify two target time-points $g = 1.5, 2.5$ which correspond to the third and fifth stages, respectively. The details of Examples 1–3 are described as follows:

Example 1. For the i th subject at the m th stage, we generate the assigned treatment A_{im} independently from a discrete Uniform $\{1, 2, 3\}$ distribution. For each $1 \leq i \leq n$ and $1 \leq m \leq 5$, we let

$$d_{im} = \operatorname{argmin}_{k \in \{1, 2, 3\}} \operatorname{Ang}\{f_m(\mathbf{H}_{im}), \mathbf{V}_k\},$$

where $f_m(\mathbf{H}_{im}) = (f_{1m}(\mathbf{H}_{im}), f_{2m}(\mathbf{H}_{im}))^\top = (\mathbf{H}_{im}^\top \boldsymbol{\theta}_{m1}, \mathbf{H}_{im}^\top \boldsymbol{\theta}_{m2})^\top$. Let $\boldsymbol{\theta}_{1j} = \boldsymbol{\theta}_j^\top$, $\boldsymbol{\theta}_{2j} = (0.5\boldsymbol{\theta}_j, -1.5\boldsymbol{\theta}_j, 0.1)^\top$, $\boldsymbol{\theta}_{3j} = (0.25\boldsymbol{\theta}_j, -0.5\boldsymbol{\theta}_j, \boldsymbol{\theta}_j, 0.05, 0.1)^\top$,

$\boldsymbol{\theta}_{4j} = (0.1\boldsymbol{\theta}_j, -0.25\boldsymbol{\theta}_j, 0.5\boldsymbol{\theta}_j, \boldsymbol{\theta}_j, 0.01, -0.05, 0.1)^\top$, and $\boldsymbol{\theta}_{5j} = (0.05\boldsymbol{\theta}_j, -0.1\boldsymbol{\theta}_j, 0.25\boldsymbol{\theta}_j, 0.5\boldsymbol{\theta}_j, \boldsymbol{\theta}_j, 0.01, -0.05, 0.05, -0.1)^\top$ for $j = 1, 2$, where $\boldsymbol{\theta}_1 = (1, 1, 1, 0, \dots, 0)$ and $\boldsymbol{\theta}_2 = (1, -1, -1, 0, \dots, 0)$ are both p -dimensional vectors. We generate covariates $\mathbf{X}_{im}$ from a multivariate normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{I}_p)$. We let $T_i = 0.5 + \sum_{m=1}^5 T_{im}$ with

$$T_{im} = \begin{cases} 0.5I(A_{im} = d_{im})W_{im} + I(A_{im} \neq d_{im})U_{im}, & \text{if } T_{i(m-1)} \geq 0.5, \\ 0, & \text{if } T_{i(m-1)} < 0.5, \end{cases}$$

where $T_{i0} = 0.5$, $W_{im} \sim \text{Bin}(1, 0.95 - 0.1I\{X_{im1} + X_{im3} > 0\})$ and $U_{im} \sim \text{Multinomial}(0.5, 0.25, 0)$ with probabilities $(0.60, 0.15, 0.25)$. Then T_i is a discrete random variable with a value from $\{0.5, 0.75, 1, 1.25, \dots, 3\}$. We generate the censoring time $C_i \sim \text{Multinomial}\{0.5, 1, 1.5, 2, 2.5, 3\}$ with certain probabilities corresponding to the censoring rate 0.38, 0.51, and 0.61, respectively.

Example 1 represents a case with linear classification functions and angle-based decision rules. Note that the survival time T_i in Example 1 satisfies Assumption 3, since for each time-point s , $I(T_i = s) = I(T_{im(s)-1} = 0.5) \cdot I(T_{im(s)} = s - 0.5m(s))$ does not depend on $\mathbf{X}_{im}$ and A_{im} for $m > m(s)$. Moreover, since T_i is discrete, $I(T_i < s) = \sum_{j=1}^{s-1} I(T_i = j)$ also does not depend on $\mathbf{X}_{im}$ and A_{im} for $m > m(s)$.

Under Example 1, the proposed method outperforms the competing methods. According to Table 1, when $g = 2.5$ with 61% censoring rate, the improvement rates of the proposed method are 66.5% and 74.3% compared to the censored Q-learning and the subgroup methods, respectively. In contrast to the subgroup method, which only incorporates patients' information from a single stage to estimate the optimal decision rule for that stage, the proposed method uses information from all stages, and can thus achieve better performance. In addition, the improvement of the proposed method compared to the censored Q-learning method is likely due to that the proposed method can directly estimate all decision rules simultaneously without modeling the relationship between the survival outcome and treatments. We present the boxplots of the estimated survival probability in Figures 3 and 4.

Example 2. This example is the same as Example 1, except that we do not generate A_{im} from a uniform distribution. In particular, we specify the true propensity score as follows. For

Table 1. Example 1: The estimated conditional survival probability $\hat{\mathcal{S}}^{\mathcal{D}}(g)$ with standard deviations (SD) in parentheses based on 150 simulations.

Stage	g	Censoring rate	Method	$\hat{\mathcal{S}}^{\mathcal{D}}(g)(\text{SD})$	Imp-rate
3	1.5	61%	Oracle	0.740(0.072)	–
			Proposed	0.655(0.084)	–
			Qlearning	0.419(0.110)	56.3%
5	2.5	61%	Subgroup	0.429(0.096)	52.7%
			Oracle	0.618(0.211)	–
			Proposed	0.483(0.214)	–
			Qlearning	0.290(0.155)	66.5%
			Subgroup	0.277(0.174)	74.3%

NOTE: "Imp-rate" represents the improvement rate of the proposed method compared to other methods.

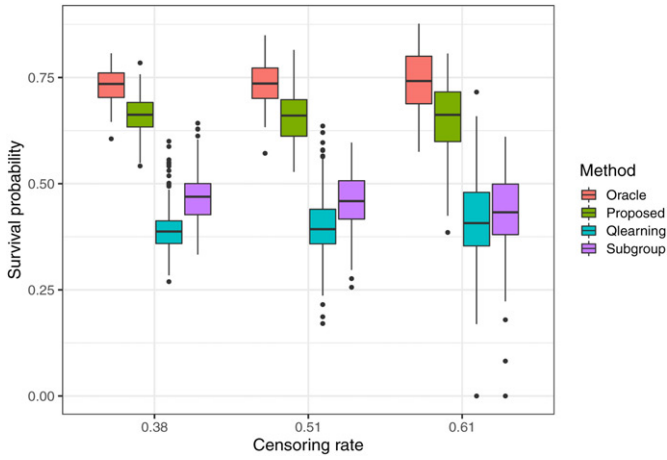


Figure 3. Boxplots of the estimated survival probabilities for the third stage of different censoring rates in Example 1.

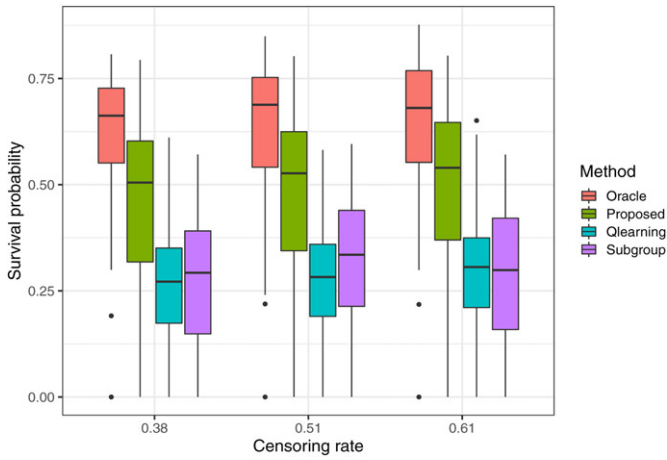


Figure 4. Boxplots of the estimated survival probabilities for the fifth stage of different censoring rates in Example 1.

each $k = 1, 2, 3$ and $m = 1, \dots, 5$,

$$P(A_{im} = k | \mathbf{H}_{im}) = \frac{\exp(\mathbf{H}_{im}^\top \boldsymbol{\gamma}_{km})}{\sum_{j=1}^3 \exp(\mathbf{H}_{im}^\top \boldsymbol{\gamma}_{jm})},$$

where $\boldsymbol{\gamma}_{jm} = (\boldsymbol{\gamma}_j, \dots, \boldsymbol{\gamma}_j, \mathbf{0}_{m-1})^\top$ consists of a $(m-1)$ -dimensional zero vector $\mathbf{0}_{m-1}$ and m copies of p -dimensional vector $\boldsymbol{\gamma}_j$ with $\boldsymbol{\gamma}_1 = (0, 0, 0.25, 0, \dots, 0)$, $\boldsymbol{\gamma}_2 = (0, -0.25, 0.5, 0, \dots, 0)$ and $\boldsymbol{\gamma}_3 = (0, 0, 0, 0, \dots, 0)$.

To mimic observational studies, the treatments are not uniformly assigned and the probabilities of treatment assignments (the propensity scores) are unknown. We use the penalized multinomial model mentioned in Section 5 to estimate the propensity scores. Table 2 indicates that the proposed method still leads to larger estimated conditional survival probabilities than the two existing methods. The improvement rates compared to the censored Q-learning and subgroup methods are about 55.6% and 43.0%, respectively, for $g = 1.5$. The boxplots of the estimated survival probabilities for the third and fifth stages under different censoring rates are provided in the supplementary materials. In addition, we conduct simulations to illustrate estimation accuracy of propensity scores, and provide the results in the supplementary materials.

Table 2. Example 2: The estimated conditional survival probability $\hat{\mathcal{S}}\hat{\mathcal{D}}(g)$ with standard deviations (SD) in parentheses based on 150 simulations.

Stage	g	Censoring rate	Method	$\hat{\mathcal{S}}\hat{\mathcal{D}}(g)(\text{SD})$	Imp-rate
3	1.5	61%	Oracle	0.737(0.080)	–
			Proposed	0.655(0.093)	–
			Qlearning	0.421(0.105)	55.6%
			Subgroup	0.458(0.099)	43.0%
5	2.5	61%	Oracle	0.648(0.199)	–
			Proposed	0.439(0.228)	–
			Qlearning	0.276(0.170)	58.7%
			Subgroup	0.273(0.170)	61.0%

NOTE: “Imp-rate” represents the improvement rate of the proposed method compared to other methods.

Table 3. Example 3: The estimated conditional survival probability $\hat{\mathcal{S}}\hat{\mathcal{D}}(g)$ with standard deviations (SD) in parentheses based on 150 simulations.

Stage	g	Censoring rate	Method	$\hat{\mathcal{S}}\hat{\mathcal{D}}(g)(\text{SD})$	Imp-rate
3	1.5	60%	Oracle	0.745(0.030)	–
			Proposed	0.648(0.031)	–
			Qlearning	0.464(0.032)	39.7%
			Subgroup	0.502(0.039)	29.1%
5	2.5	60%	DWSurv	0.490(0.036)	32.4%
			Oracle	0.461(0.103)	–
			Proposed	0.365(0.111)	–
			Qlearning	0.249(0.093)	46.6%
			Subgroup	0.265(0.101)	37.7%
			DWSurv	0.270(0.147)	35.2%

NOTE: “Imp-rate” represents the improvement rate of the proposed method compared to other methods.

Example 3. This example is the same as Example 1, except that we consider binary treatment choices and change the distributions of W_{im} and U_{im} . For the i th subject at the m th stage, we let $d_{im} = \text{sign}\{f_m(\mathbf{H}_{im})\}$, where $f_m(\mathbf{H}_{im}) = \mathbf{H}_{im}^\top \boldsymbol{\theta}_m$, $\boldsymbol{\theta}_1 = \boldsymbol{\theta}_0^\top$, $\boldsymbol{\theta}_2 = (0.5\boldsymbol{\theta}_0, -1.5\boldsymbol{\theta}_0, 0.1)^\top$, $\boldsymbol{\theta}_3 = (0.25\boldsymbol{\theta}_0, -0.5\boldsymbol{\theta}_0, 0.05, 0.1)^\top$, $\boldsymbol{\theta}_4 = (0.1\boldsymbol{\theta}_0, -0.25\boldsymbol{\theta}_0, 0.5\boldsymbol{\theta}_0, 0.01, -0.05, 0.1)^\top$, and $\boldsymbol{\theta}_5 = (0.05\boldsymbol{\theta}_0, -0.1\boldsymbol{\theta}_0, 0.25\boldsymbol{\theta}_0, 0.5\boldsymbol{\theta}_0, 0.01, -0.05, 0.05, -0.1)^\top$ with a p -dimensional vector $\boldsymbol{\theta}_0 = (1, 1, 1, 0, \dots, 0)$. We generate A_{im} independently from a discrete Uniform $\{1, -1\}$ distribution. In addition, we let $W_{im} \sim \text{Multinomial}(0.5, 0.25)$ with probabilities $\{0.85 - 0.05I(X_{im1}^3 + X_{im3} > 0), 0.15 + 0.05I(X_{im1}^3 + X_{im3} \leq 0)\}$ and $U_{im} \sim \text{Multinomial}(0.5, 0.25, 0)$ with probabilities $\{0.6 - 0.025I(X_{im5}^2 - X_{im6} < 0.1), 0.25 + 0.025I(X_{im5}^2 - X_{im6} \geq 0.1), 0.15\}$. We generate the censoring time $C_i \sim \text{Multinomial}\{0.5, 1, 1.5, 2, 2.5, 3\}$ with probabilities corresponding to the different censoring rate, respectively.

In this example, we restrict the proposed method to a binary treatment setting. To conduct a more comprehensive numerical comparison, we include the DWSurv method as an additional competing method. Table 3 shows that the proposed method still achieves the best performance among all the methods. In particular, when we evaluate the survival probability at the fifth stage, the proposed method has 46.6%, 37.7%, and 35.2% improved the conditional survival probabilities compared to the Q-learning, Subgroup, and DWSurv, respectively. The boxplots of the estimated survival probabilities for the third and fifth stages under different censoring rates are provided in the supplementary materials.

Overall, the proposed method can achieve better numerical performance than existing methods and estimate the optimal DTR more accurately under a high censoring rate. One possible reason is that the proposed method uses more information via incorporating patients' information from all stages instead of from only one single stage. In addition, the proposed method avoids modeling the relationship between the survival time and treatments or covariates, and estimates the optimal DTR directly. In contrast, the censored Q-learning and DWSurv methods require to model the relationship between the survival time and treatments, which may cause model misspecification in multistage settings.

7. Real Data Examples

In this section, we apply the proposed method to the Framingham heart study data and AIDS clinical data for assessing its performance. The proposed method is implemented with linear classification functions. For comparison, we consider the censored Q-learning method (Goldberg and Kosorok 2012) and the subgroup identification method (Huling and Yu 2018) with one-versus-rest extension, the same as in Section 6. In Section 7.1 on binary treatment setting, we also include the DWSurv (Simoneau et al. 2020) method for comparison.

7.1. Application to Framingham Heart Study Data

The Framingham heart study, established in 1948, is the first longitudinal prospective large-scale cohort to study cardiovascular disease in the United States. The epidemiological and genetic risk factors for cardiovascular disease are investigated by monitoring original cohort participants. According to Tsao and Vasan (2015), up to 32 examinations are performed for each individual biannually during the 65 years of follow-up. Here we only use participants' information from the second visit to the sixth visit, where each visit represents one stage, and different participants can have different numbers of visits. Typically, an antihypertensive treatment will be recommended to individual participants based on the level of blood pressure. Nevertheless, the accuracy of recommendation may be potentially improved if we take more information about participants into account, so that the overall survival probability of cardiovascular disease could be improved. In this study, we aim to maximize the overall survival probability of cardiovascular disease by customizing sequential decision rules which inform each participant whether an antihypertensive treatment should be taken to balance the blood pressure level at each decision point.

Specifically, we obtain participants' observed survival time (Y_i), death reviewed status (Δ_i), visit time, and whether to take the antihypertensive treatment ($A_{im} = 0$ or 1) at each visit m from the data. In our analysis, we incorporate the following information as covariates (X_{im}) collected at each visit: age, cholesterol, high-density lipoprotein, diastolic blood pressure, and the presence of diabetes. There are a total of 3939 participants who have completely observed covariates at each visit before Y_i , that is, $n = 3939$. The death event rate is 22.9% by the end of the study. We randomly select 1000 participants as a

Table 4. The estimated conditional survival probability $\hat{\xi}^{\mathcal{D}}(g)$ with standard deviations (SD) in parentheses based on 50 repetitions.

Stage	g	Method	$\hat{\xi}^{\mathcal{D}}(g)$ (SD)	Imp-rate
3	4775	Proposed	0.955(0.007)	–
		Qlearning	0.894(0.008)	6.4%
		Subgroup	0.932(0.004)	2.4%
		DWSurv	0.930(0.008)	2.6%
4	5946	Proposed	0.926(0.014)	–
		Qlearning	0.884(0.008)	4.5%
		Subgroup	0.896(0.007)	3.2%
		DWSurv	0.885(0.013)	4.4%
5	7376	Proposed	0.913(0.023)	–
		Qlearning	0.706(0.026)	22.7%
		Subgroup	0.820(0.012)	10.2%
		DWSurv	0.805(0.017)	11.8%
6	8539	Proposed	0.823(0.008)	–
		Qlearning	0.575(0.029)	30.1%
		Subgroup	0.748(0.017)	9.1%
		DWSurv	0.738(0.022)	10.3%

NOTE: "Imp-rate" represents the improvement rate of the proposed method compared to other methods.

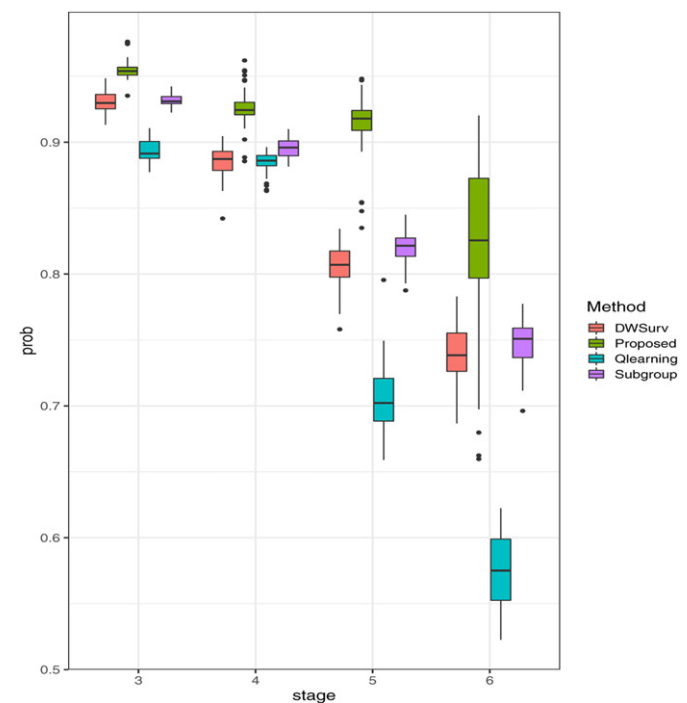


Figure 5. Boxplot of the estimated conditional survival probabilities for Framingham heart study data.

training set and the other 1000 participants as a testing set for 50 iterations. We consider four target time-points $g = 4775, 5946, 7376$, and 8539 , and calculate the estimated conditional survival probability at each g for each testing set given the recommended sequential decision rules estimated on the corresponding training set.

The average values of the estimated conditional survival probabilities and the corresponding boxplots are presented in Table 4 and Figure 5, respectively. Table 4 and Figure 5 show that the proposed method achieves the highest survival probabilities at all different target time points. The improvements of the proposed method compared to the censored Q-learning are at least 4.5%, and the improvements of the proposed method compared to the Subgroup and the DWSurv are at least 2.4%.

Notice that as the number of stages increases, the proposed method improves even more than the competing methods. This is possibly due to the fact that the proposed method integrates information from all stages into one single algorithm, which is more effective than the methods only considering information from one single stage information at a time. In particular, the improvement of the proposed method compared to the censored Q-learning achieves 30.1% at the final stage. Note that the censored Q-learning requires to model relationships between the survival time and treatments, which likely leads to model misspecification in multistage settings.

7.2. Application to AIDS Clinical Data

In this subsection, we apply the proposed method to data from the AIDS Clinical Trials Group (ACTG) 175 (Hammer et al. 1996). ACTG 175 is a double-blind and randomized clinical trial, which uses the counts of CD4⁺ T cells of patients to compare monotherapies and combination therapies of HIV infection. In this trial, 2139 patients ($n = 2139$) with HIV infection were randomly assigned to four treatment groups ($K = 4$) with the same probability: zidovudine (ZDV) monotherapy, ZDV and didanosine (ddI), ZDV and zalcitabine (ZAL), and ddI monotherapy. We obtain information regarding whether events happen or not for a patient (Δ_i), and the observed time (Y_i) when patients are censored or events occur. Here, an event refers to the patient's death, a decline of CD4⁺ T cell counts being at least 50, or an event indicating progression to AIDS. In addition to the treatment and event time, we consider 12 clinical covariates ($p = 12$) in our analysis, which are also included in Fan et al. (2017) and Qi and Liu (2018). Five of the 12 covariates are continuous: weight, CD4⁺ T cells counts at baseline, CD8 count at baseline, age, and Karnofsky score, where Karnofsky score is a way to rate a person's ability to perform activities of daily living, and ranges from 0 to 100, with a higher score indicating that a person is more able to perform daily activities. The remaining seven covariates are binary: gender (0 = female, 1 = male), race (0 = white, 1 = non-white), homosexual activity (0 = no, 1 = yes), history of intravenous drug use (0 = no, 1 = yes), symptomatic status (0 = asymptomatic, 1 = symptomatic), antiretroviral history (0 = naive, 1 = experienced), and hemophilia (0 = no, 1 = yes).

This study only involves one decision point (one stage), but with multicategory treatments which contain four different treatment choices. Similarly as in Section 7.1, we randomly select 1000 participants as a training set and the other 1000 participants as a testing set for 50 iterations. We specify three target time-points $g = 861, 984$, and 1169 , and calculate the estimated conditional survival probability at each g for each testing set given recommended sequential decision rules estimated on the corresponding training set.

Table 5 and Figure 6 show that the proposed method achieves the highest survival probabilities at all target time points. In addition, as the g increases, the proposed method performs better compared to competing methods. Note that the censored Q-learning performs slightly better than Subgroup on this dataset, as the censored Q-learning can handle multicategory treatments but Subgroup is only designed for binary treatments. Although

Table 5. The estimated conditional survival probability $\hat{S}^{\mathcal{D}}(g)$ with standard deviations (SD) with 50 repetitions.

g	Method	$\hat{S}^{\mathcal{D}}(g)(SD)$	Imp-rate
861	Proposed	0.769(0.049)	—
	Qlearning	0.761(0.059)	1.04%
	Subgroup	0.732(0.078)	5.05%
984	Proposed	0.714(0.048)	—
	Qlearning	0.705(0.063)	1.26%
	Subgroup	0.683(0.075)	4.54%
1169	Proposed	0.683(0.051)	—
	Qlearning	0.665(0.061)	2.64%
	Subgroup	0.657(0.071)	3.96%

NOTE: "Imp-rate" represents the improvement rate of the proposed method compared to other methods.

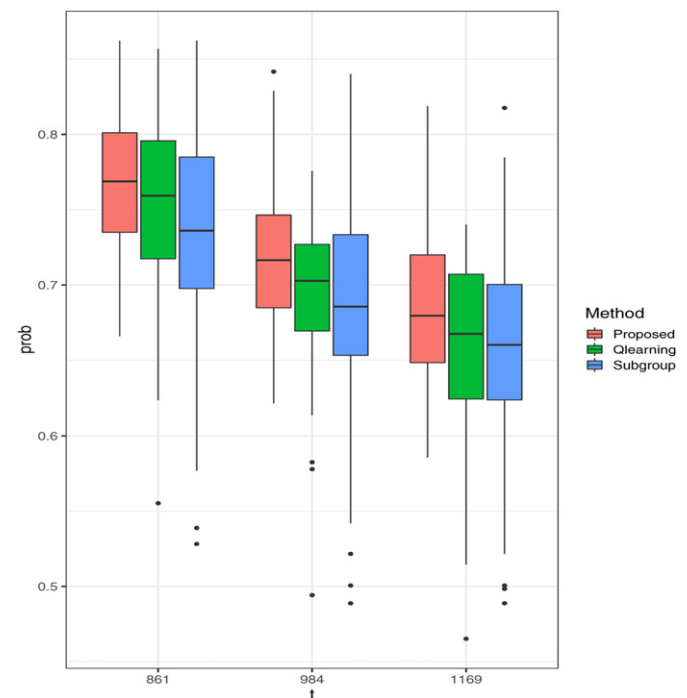


Figure 6. Boxplot of the estimated conditional survival probabilities in AIDS clinical data.

there is only one stage for treatment decision, the higher survival probabilities of the proposed method still indicate that patients would receive more benefits if the proposed strategy is applied.

8. Discussion

In this article, we propose a novel weighted KM estimator to estimate the survival function for patients following a specific multicategory DTR, and propose an objective function under an angle-based framework for estimating the optimal DTR. We formulate the estimation of decision rules at multiple stages as one unconstrained optimization problem. In this way, the proposed method is more computationally efficient than the multicategory methods with the sum-to-zero constraint. Also, the proposed method avoids modeling the relationship between the survival time and covariates or treatments. This circumvents a potential model-misspecification issue which raises in some existing methods such as the censored Q-learning and DWSurv.

Although the proposed optimization problem (24) is non-convex, we use different initial values to find a solution close to an optimal value in our implementation. In future work, we may improve our implementation via adopting more advanced methods to achieve a near-optimal solution for nonconvex optimization. For instance, Li and Fukushima (2001) modified the BFGS method in nonconvex minimization for obtaining the global convergence property. Liu, Shen, and Doss (2005) and Liu and Shen (2006) solved the nonconvex optimization problem by minimizing a sequence of convex upper approximations of the objective function. Chen et al. (2018) considered gradient descent with random initialization.

Several other improvements and extensions are worth exploring in the future. Although the proposed method is implemented for a linear classification function, it is also feasible to extend the proposed method to estimate nonlinear classification functions by applying Gaussian kernel learning classifiers. In addition, the goal of the proposed method is to maximize conditional survival probability, which can also be generalized to target maximizing other outcomes of interests, for example, mean residual life and median survival time. It is also meaningful to extend the proposed method for continuous treatments, such as drug dosages. Moreover, developing statistical inference for prediction is useful in practice, for example, generalizing our method to quantify uncertainty via confidence interval and hypothesis testing.

Supplementary Materials

The supplementary materials contain additional simulation results, a proposition for scenarios where Condition 4 holds, and the proofs of Theorems 1 and 2.

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